Continuous Assessment 3 (CA-3) - Practice Questions

Course Code: CSE252

Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Section(s): K3P25RB

Syllabus Covered: Unit 3 to Unit 5

CA-3 pattern: 3 questions x 10 marks = 30

One question from each of Units 3, 4 and 5

Instructions: prepare every question below. Each carries 10 marks; the marks for each part are shown in brackets. Write structured answers, show all working for numerical parts, and draw diagrams wherever they help.

Unit 3: Supervised Machine Learning

1. A school uses machine learning to predict whether each student will pass or fail a course. (10 Marks)

(a) Describe how a decision tree reaches a prediction for a new record. [5]

(b) Describe how K-Nearest Neighbour classifies a new record and what happens when K is very small or very large. If the three nearest neighbours of a new student are labelled Pass, Fail and Pass, state the label that 3-Nearest Neighbour predicts. [5]

2. A decision tree that predicts whether a loan applicant will repay must choose which feature to split on at every node. (10 Marks)

(a) Explain what entropy and Gini impurity measure at a node, and state the value of each for a pure node and for a 50/50 two-class node. [5]

(b) Explain information gain, and how the tree uses it to choose a split. [5]

3. A coaching institute wants to predict a student's test marks from the number of hours the student studies each week. (10 Marks)

(a) Explain what simple linear regression fits to such data and what quantity it minimises. [5]

(b) Explain what the slope and the intercept of the fitted line each represent in this setting, and state one sign that a straight line is not a good fit to the data, with the reason. [5]

4. A data scientist trains several models of different complexity on the same student-performance dataset and compares their training and test results. (10 Marks)

(a) Explain what underfitting and overfitting mean, linking each to bias and variance. [5]

(b) Describe how the training and test performance of a model show whether it is underfitting, overfitting or well fitted, and suggest one change that would improve an overfitting model and one that would improve an underfitting model. [5]

5. Logistic regression and Naive Bayes are two probabilistic classifiers that are widely used for yes / no decisions. (10 Marks)

(a) Describe how logistic regression turns a score into a probability using the sigmoid function, and give one application in which it is a good choice, with a reason. [5]

(b) Explain why Naive Bayes is called "naive", and give one application in which it is a good choice, with a reason. [5]

Unit 4: Unsupervised Learning

6. A university library groups its books by the borrowing patterns of their readers using hierarchical clustering and shows the result as a dendrogram. (10 Marks)

(a) Explain how hierarchical clustering builds its clusters, from single records up to one cluster, and why it does not need k in advance. [5]

(b) Distinguish single linkage from complete linkage, stating the kind of clusters each tends to produce, and explain how the number of clusters is chosen from a dendrogram. [5]

7. An online store wants to divide its customers into groups by running k-Means on their annual spend and their number of orders. (10 Marks)

(a) Explain the steps of the k-Means algorithm. [5]

(b) Explain how the elbow method is used to choose the value of k, and state three limitations of k-Means. [5]

8. A supermarket studies its bills to find products that are often bought together (market basket analysis). (10 Marks)

(a) Define support, confidence and lift, and state what each one tells the store manager. [5]

(b) Explain the Apriori principle and how it reduces the work of finding frequent itemsets, and give one action the supermarket could take on finding a strong rule such as {bread} -> {butter}. [5]

9. A factory records 30 sensor readings for every machine and wants to reduce them to a few features with Principal Component Analysis. (10 Marks)

(a) Describe the main steps of Principal Component Analysis. [5]

(b) State two reasons why reducing the number of features is useful, and explain how the explained-variance ratio is used to decide how many components to keep. [5]

10. A delivery company records GPS points along curved city roads, together with a few faulty readings, and must choose between k-Means and DBSCAN to group them. (10 Marks)

(a) Explain what core points, border points and noise points are in DBSCAN. [5]

(b) Compare k-Means and DBSCAN on the shape of the clusters each can find, whether the number of clusters must be given, and how each treats outliers, and recommend one of the two for the GPS data, with the reason. [5]

Unit 5: Model Evaluation and Improvement

11. A model predicts the delivery times of four parcels. Its errors (actual minus predicted, in minutes) are 2, -2, 4 and -4. (10 Marks)

(a) Define MAE, MSE and RMSE, and state why RMSE is often reported instead of MSE. [5]

(b) Compute the MAE, the MSE and the RMSE of these predictions, showing the working. [5]

12. A decision tree classifier is tuned on a dataset of 1,000 rows using 5-fold cross validation and grid search. (10 Marks)

(a) Explain k-fold cross validation and why it is better than a single train-test split, and state how many rows are used for training and how many for validation in each fold. [5]

(b) The five folds score 0.80, 0.84, 0.78, 0.82 and 0.86, and the grid tries 3 values of the maximum depth of the tree and 4 values of the minimum number of samples in a leaf. Compute the average accuracy, the number of models trained, and the total time if one model takes 10 seconds to train. [5]

13. A disease-screening model is tested on 200 patients, 50 of whom have the disease. At a threshold of 0.5 it flags 40 patients, of whom 30 have the disease. When the threshold is lowered to 0.3 it flags 70 patients, of whom 45 have the disease. (10 Marks)

(a) Compute the precision and the recall at each threshold, showing the working. [5]

(b) State which threshold you would choose for screening, with the reason, and explain what a ROC-AUC of 0.5 means. [5]

14. A spam filter is tested on 100 emails, of which 40 are spam. The filter marks 30 emails as spam, and 25 of those are really spam. (10 Marks)

(a) Construct the confusion matrix, and compute the accuracy. [5]

(b) Compute the precision and the recall, showing the working, and state which of the two matters more for a spam filter, with the reason. [5]

15. A fraud detector correctly catches 40 fraudulent transactions, raises 10 false alarms on genuine transactions, and misses 20 frauds. (10 Marks)

(a) Define precision, recall and the F1 score using TP, FP and FN, and give one example where recall matters more than precision, with the reason. [5]

(b) Compute the precision, the recall and the F1 score of the detector, showing the working. [5]

----- End of Practice Questions -----